Semantic Mapping of Psychological Assessment Questions to BERTopic-Derived Solitude-Related Concepts Using SciBERT Embeddings

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Abstract

Background:
This study is part of the Promoting Healthy Aging through Semantic Enrichment of Solitude Research (PHASES) project, which aims to develop a structured ontology of psychological constructs related to solitude and gerotranscendence. In our previous work, BERTopic modeling [1] was applied to *The Handbook of Solitude: Psychological Perspectives on Social Isolation, Social Withdrawal, and Being Alone* to extract topics, keywords, topic labels, and topic summaries. Solitude is characterized by a dominant relationship with the self and by physical or mental separation from immediate social demands. This study extends that work by semantically linking psychological assessment questions to BERTopic-derived topic representations using SciBERT embeddings. Assessment questions were drawn from established instruments, including the UCLA Loneliness Scale (Versions 1, 2, and 3), Differential Loneliness Scale (DLS), Inventory of Interpersonal Problems (IIP), and Subjective Happiness and Loneliness Scales.

Methods:
Topic summaries and labels generated from the previous BERTopic analysis were loaded from a structured dataset. Assessment questions were extracted from a separate text file. The assessment questions and BERTopic-derived topic summaries were independently encoded using SciBERT, a transformer-based language model pretrained on scientific literature to generate contextual embeddings. Cosine similarity was computed between each question embedding and all topic-summary embeddings to measure semantic similarity.

Results:
The proposed retrieval framework successfully identified, for each assessment question from multiple psychological instruments, the five most semantically similar BERTopic-derived topic summaries together with their corresponding topic labels. For example, for the question *"I am unhappy doing so many things alone,"* the highest-ranked topic was *Solitude* (0.652), followed by *Creativity* (0.648), *Primates* (0.646), *Smartphone* (0.644), and *Lonely* (0.643). However, because the topic summaries encompass concepts related to being alone, independent activity, social isolation, technology-related loneliness, and loneliness-related experiences, some highly ranked topics may not be directly relevant to solitude. Instead, they may reflect broader semantic associations captured by the embeddings and therefore require validation by subject matter experts.

Conclusion:
The proposed framework demonstrates the effectiveness of SciBERT-based semantic matching for linking psychological assessment questions to topic representations derived from prior BERTopic analysis. By leveraging contextual embeddings and cosine similarity, the approach captures semantic relationships beyond keyword overlap and provides interpretable semantic matches between assessment questions and BERTopic-derived topic representations. The framework facilitates the identification of candidate psychological constructs relevant to solitude and ontology development. However, subject matter expert review is necessary to distinguish genuinely relevant concepts from semantically related but conceptually irrelevant topics.

Keywords: SciBERT; BERTopic; semantic similarity; psychological assessment; solitude; ontology development.

Project Website: http://buffalo-ontology-group.github.io/phases-healthy-aging.github.io/
Source Code: https://github.com/Buffalo-Ontology-Group/phases-nlp
License: MIT License

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